

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and shipping large-scale distributed systems. Built **Go**-based streaming infrastructure at YouTube handling billions of events daily. Architected **Kubernetes**, **gRPC**, and **Kafka** pipelines at Whatnot supporting real-time marketplace functionality for millions of users. Led teams shipping **Spanner**-powered search infrastructure serving **100+ million daily active users** at Google. Partnered with ML, product, and infrastructure teams to deliver ads ranking systems and recommendation pipelines. Focuses on building observable, maintainable systems with strong on-call practices. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, SQL

Distributed Systems: Go, Kubernetes, gRPC, Kafka, Spanner, Bigtable, MapReduce, Pub/Sub

Infrastructure: Linux, networking fundamentals, containerization, service mesh, CI/CD pipelines, disaster recovery

Data Engineering: data pipelines, stream processing, schema design, ETL systems, data warehousing

Leadership: technical strategy, team management, cross-functional collaboration, systems design, on-call leadership

Work Experience

YouTube

Staff Software Engineer | 02/2025 - Present

- Architected Go-based streaming infrastructure processing billions of events daily, implementing fault-tolerant pipeline design with sub-second latency requirements.
- Designed real-time analytics systems using Kubernetes orchestration, reducing infrastructure costs by 20% while maintaining 99.9% availability targets.
- Led technical design reviews for new system migrations, ensuring backward compatibility and minimal user impact during rollouts.
- Mentored engineers on Go best practices and distributed systems patterns for team-wide skill development.
- Established observability standards including distributed tracing and SLO dashboards across team services.

Whatnot

Staff Software Engineer | 05/2024 - 02/2025

- Built real-time marketplace platform using Go and Kubernetes, supporting live video shopping experiences for hundreds of thousands of concurrent users.
- Implemented gRPC service architecture and Kafka event streaming for order processing, reducing checkout latency by 40%.
- Designed multi-region deployment strategy with disaster recovery automation, achieving RTO under 15 minutes for critical services.
- Established on-call rotation and incident response processes, improving MTTR from 45 minutes to under 10 minutes.

- Led technical roadmap planning and vendor evaluation for infrastructure tooling investments.

Google

Staff Software Engineer/Manager | 03/2015 - 04/2024

- Led Search infrastructure team serving over **100 million daily active users**, architecting Spanner-based storage systems achieving 99.99% availability.
- Managed team of 8 engineers delivering ads ranking infrastructure processing millions of queries per second with sub-50ms p99 latency.
- Migrated legacy MapReduce pipelines to Bigtable-backed services, reducing processing latency from hours to minutes for critical reporting systems.
- Defined and enforced engineering standards for code review, testing coverage, and deployment procedures across multiple teams.
- Conducted hiring interviews and onboarding programs, scaling team from 4 to 12 engineers over 3 years.

Google

Software Engineer | 02/2011 - 02/2015

- Developed backend services for ad targeting systems using C++ and Java, improving click-through rate measurement accuracy by 15%.
- Optimized query performance for internal reporting tools, reducing average response time from 8 seconds to under 1 second.
- Built data pipeline infrastructure processing terabytes of logs daily for campaign analytics and reporting.
- Participated in on-call rotation for production systems, triaging and resolving incidents affecting advertiser dashboards.

Microsoft

Software Design Engineer II | 10/2007 - 12/2011

- Developed test automation frameworks for Windows infrastructure services, increasing test coverage from 60% to 92%.
- Collaborated with development teams to define system requirements and acceptance criteria for new features.
- Authored integration tests for cloud storage APIs, identifying critical bugs before production deployment.
- Conducted root cause analysis for customer-reported issues, delivering permanent fixes within SLA deadlines.

Education

B.S. Honors Degrees in Computer Science

Virginia Tech | 2004 - 2007